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Figure 12.10: Parasitic Mode of Nutrition

ii. **Partial root parasite**: Example: *Santalum album* (Sandal wood tree) in its juvenile stage produces haustoria which grows on roots of many plants (Figure 12.10).

12.9.3 Symbiotic mode of Nutrition

a. **Lichens**: It is a mutual association of Algae and Fungi. Algae prepares food and fungi absorbs water and provides thallus structure.

b. **Mycorrhizae**: Fungi associated with roots of higher plants including Gymnosperms. Example: *Pinus*.

c. ***Rhizobium* and Legumes**: This symbiotic association fixes atmospheric nitrogen

d. **Cyanobacteria and Coralloid Roots**: This association is found in *Cycas* where

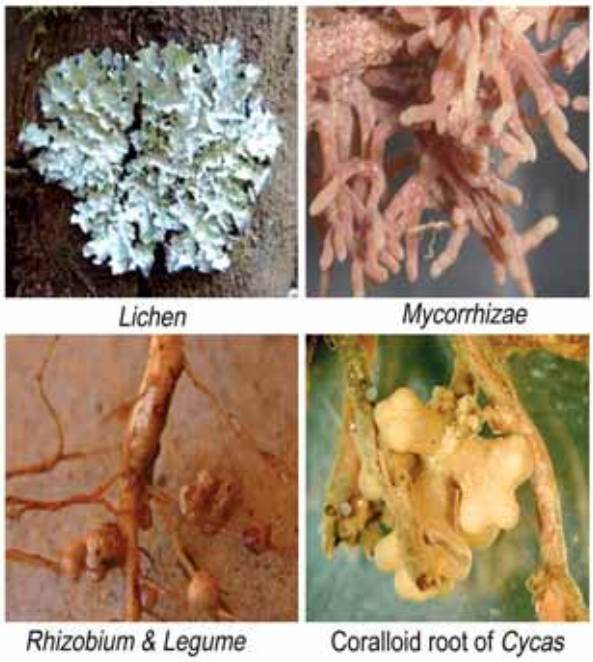


Figure 12.11: Symbiotic mode of nutrition

Nostoc associates with its coralloid roots. (Figure 12.11).

12.9.4 Insectivorous mode of nutrition

Plants which are growing in nitrogen deficient areas develop insectivorous habit to resolve nitrogen deficiency.

a. ***Nepenthes*** (Pitcher plant): Pitcher is a modified leaf and contains digestive enzymes. Rim of the pitcher is provided with nectar glands and acts as an attractive lid. When insect is trapped, proteolytic enzymes will digest the insect.

b. ***Drosera*** (Sundew): It consists of long club shaped tentacles which secrete sticky digestive fluid which looks like a sundew.

c. ***Utricularia*** (Bladder wort): Submerged plant in which leaf is modified into a bladder to collect insect in water.

d. ***Dionaea*** (Venus fly trap): Leaf of this plant modified into a colourful trap. Two folds of lamina consist of sensitive trigger hairs and when insects touch the hairs it will close (Figure 12.12).